

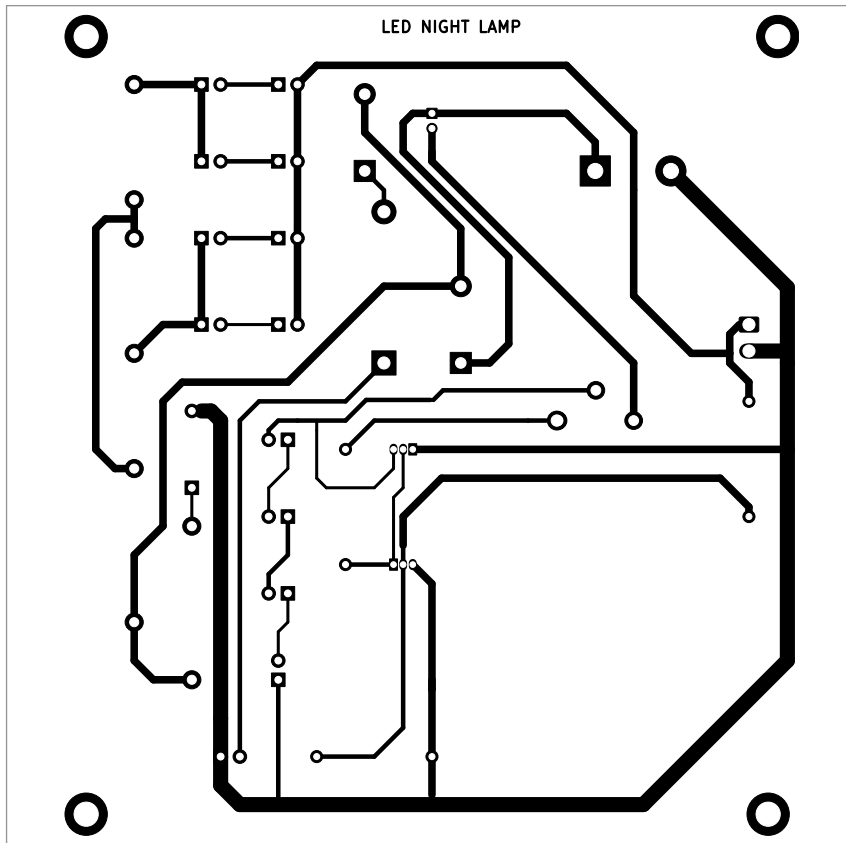


Fig. 4: Actual-size PCB layout of the circuit

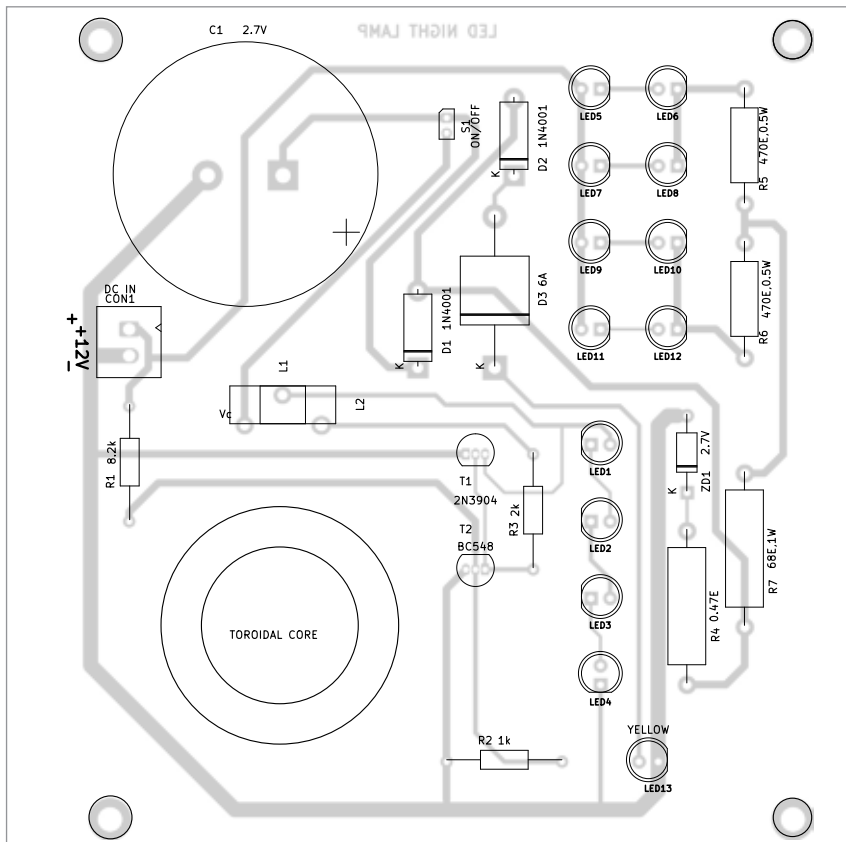


Fig. 5: Component layout of the PCB

PARTS LIST

Semiconductors:

- T1 - 2N3904 NPN transistor
- T2 - BC548 NPN transistor
- LED1-LED4 - 5mm red LED
- LED5-LED12 - 5mm green LED
- LED13 - 3mm yellow LED
- D1, D2 - 1N4001 rectifier diode
- D3 - 6-ampere power diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1 - 8.2-kilo-ohm
- R2 - 1-kilo-ohm
- R3 - 2-kilo-ohm
- R4 - 0.47-ohm, 1W
- R5, R6 - 470-ohm, 0.5W
- R7 - 68-ohm, 1W

Capacitors:

- C1 - 2.7V, 100F/500F supercapacitor

Miscellaneous:

- CON1 - 2-pin Berg male connector
- S1 - On/off toggle switch
- L1, L2 - 50 turns each of 24 SWG enameled copper wire on ferrite-core toroidal transformer
- 12V DC adaptor or power supply
- Jumper wires

(LED1 through LED13) are mounted separately on a general-purpose PCB, with yellow LED (LED13) in the middle, four red LEDs (LED1 through LED4) forming the inner LEDs, and the green LEDs (LED5 through LED12) connected in series forming the outer LEDs. However, you can place the LEDs any way you like. You can also replace the green and red LEDs with white LEDs without changing the circuitry.

On connecting 12V DC source to CON1, the green LEDs will glow and supercapacitor C1 will start charging. Initially, yellow LED13 will not glow. It will glow only when supercapacitor has been charged. Depending on the current rating of 12V DC source, the charging can take ten to twenty hours.

When the 12V DC is disconnected, or the mains power is cut off, red LEDs (LED1 through LED4) glow as night lamp. These red LEDs can remain on for days, till the capacitor voltage drops to 2V or the circuit is switched off through on/off switch S1. You can measure the capacitor voltage at test point TP1 with a voltmeter. **EFY**

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